

# 孫正義育英財団

Generation 9

ADAM CHEONG

BORN IN 2009

[READ MORE](#)

#robotics  
#engineering  
#mathematics



# Masason Foundation



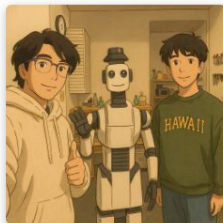
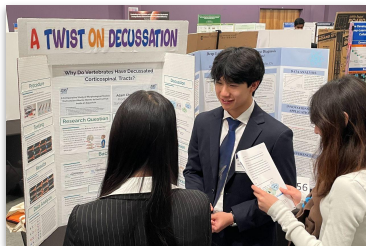
# Adam Cheong

Proof School | San Francisco, California | [adam@cheong.net](mailto:adam@cheong.net)

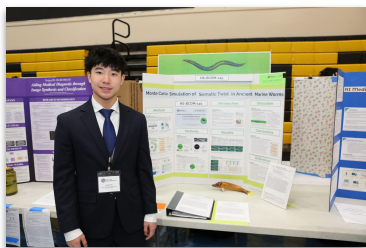
A high school sophomore from the San Francisco Bay Area with a strong interest in science, engineering, math and finance, Adam interns afterschool at a [robotics startup](#) in San Francisco. Adam aspires to identify and formulate the most important unsolved problems in robotics for his generation, and to create a future in which collective intelligence and resources are mobilized to solve them.

Adam practices bouldering, a unique skill that combines puzzle solving with physical feats — exercising both brain and brawn — as the climber places one foothold after another to chart a navigable course. From mechanical engineering to computational biology, Adam's [projects](#) since 7th grade have won STEM fair awards every year.

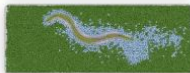
Adam is taking a [computer science class](#) this summer at UC Berkeley as a pre-college scholar, joining his older brother there. Adam plans to earn a [pilot license](#) before heading out to college, where he will double major in engineering and finance.







## Accepted Abstract



### Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms

Adam Cheong

*25th International Worm Meeting, UC Davis, Jun 2025*

[ABS](#)[PDF](#)

Monte Carlo simulation to recreate evolutionary events inside a computer in order to observe how the first digital worm organism evolves to become the first fish when given bountiful food sources.

## Submitted Paper



### Why Do Vertebrates Have Decussated Corticospinal Tracts?

Adam Cheong

*Junior Science & Humanities Symposium, San Francisco State University, Feb 2025*

[ABS](#)[PDF](#)

A comparative study of morphological models from ancient marine worms to the first fish inside an aquarium.

## Submitted Essay



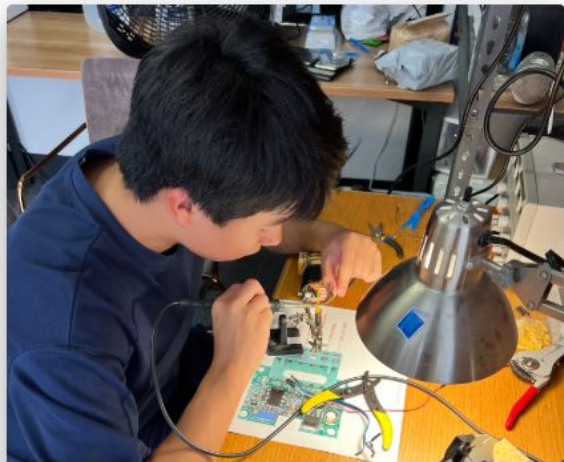
### The Rise of Homo Roboticus

Adam Cheong

*Masason Foundation, Jan 2025*

[ABS](#)[PDF](#)

10 years later, how will artificial intelligence be further developed and what kind of changes will it bring to the world?



## watney robotics

autonomous physical infrastructure



## fold,fold,fold!

dynamic duo working night shifts



## hydroponics

set up vertical farming @ Berkeley

# watney robotics

autonomous physical infrastructure

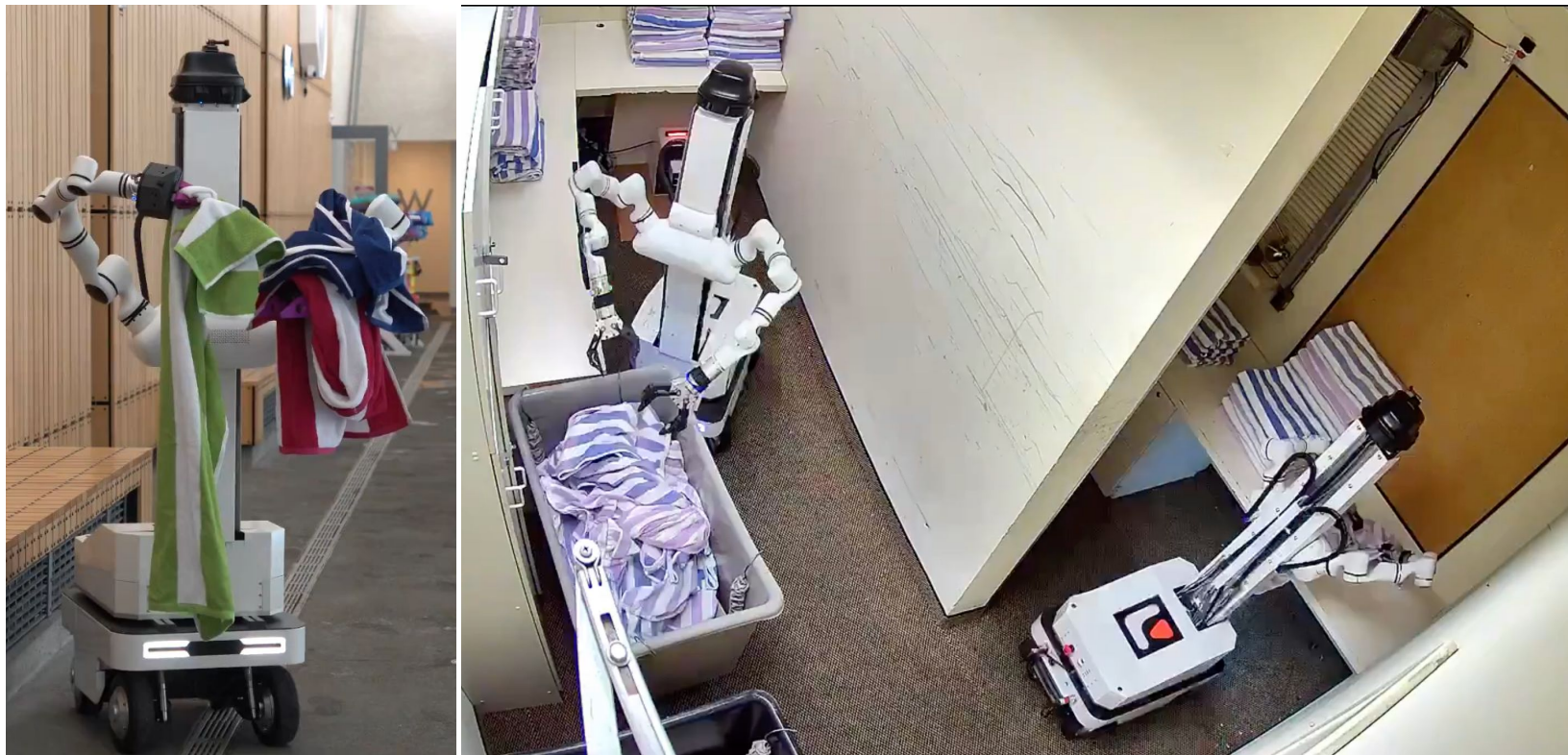


My brother's big adventure in embodied AI — so glad we could be on this quest together!



Watney alive: goes to work, cleans window, and goes shopping (circa July–Aug 2024).





Watney making itself useful poolside and in the laundry room folding towels (circa Aug–Nov 2024).



(left) Meet Watney's creator my eldest brother (circa April 2025); (right) our new offices in the SF Mission District (circa Jun 2024).





【[LINK](#) 🗣️】 Closing an ultra low-latency teleoperation loop at our Berkeley offices on Telegraph Avenue (circa Feb 14, 2024).





🤖 Watney continuously learned to become a better 🍅 hydroponics farmer (circa Nov–Dec 2023).



Our zero-to-one journey: starting 🍓 hydroponics in Berkeley (circa Aug–Oct 2023).



Heavy lifting all the way: from San Carlos home garage to new offices in Berkeley (circa July 2023).





"You can't connect the dots looking forward; you can only connect them looking backwards. So you have to trust that the dots will somehow connect in your future." — Steve Jobs (2005 Stanford commencement address).



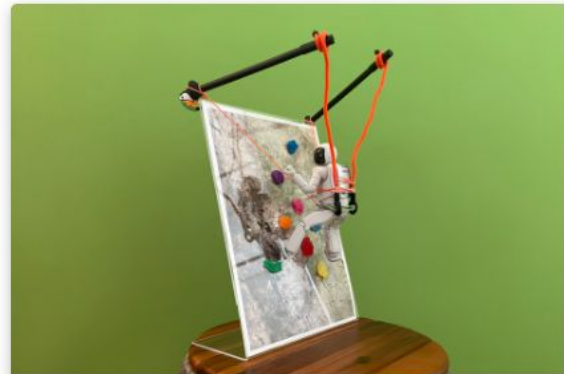
## monte carlo

Monte Carlo simulation of somatic twist in ancient marine worms



## decussation

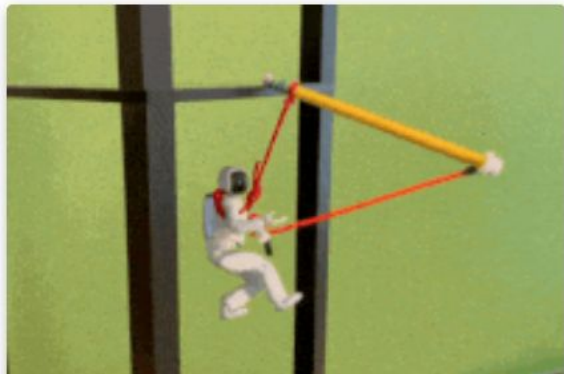
why do vertebrates have decussated corticospinal tracts?



## cliffhanger v2

dynamically stable mechanical mountain climbing aid by design





## cliffhanger

dynamically stable mechanical  
mountain climbing aid by design



## rocket fuel

homemade rocket fuel with kitchen  
chemistry (i.e.,  $\text{kno}_3$  + sugar etc.)



## go kart go!

## General

[Education](#)[Experience](#)[Volunteering](#)[Projects](#)[Science fairs](#)[Contests](#)[Posters](#)[Papers](#)[Courses](#)[Societies & clubs](#)[Skills](#)[Languages](#)[Assessments](#)

## CV



### General

**Name** Adam Cheong

**Goal** To identify and formulate the most important unsolved problems in robotics for our generation, and to create a future in which collective intelligence and resources are mobilized to solve them.

**Email** [adam@cheong.net](mailto:adam@cheong.net)

**Phone** (650) 593-8322

**Url** <https://adamcheong.com>

**Summary** Inventive high school student from San Francisco with a strong interest in science, engineering, math and finance; winner of annual STEM fair awards; natural born athlete who loves bouldering and plays the electric guitar.



General

Education

Experience

Volunteering

Projects

Science fairs

Contests

Posters

Papers

Courses

Societies &amp; clubs

Skills

Languages

Assessments

## Education

2023.08 - 2027.06

## High School Diploma

San Francisco, California



Proof School

9th to 12th Grade

- 10th: Literary Arts: Graphic Narrative
- 10th: AP Calculus BC
- 10th: AP Physics C: Mechanics
- 10th: AP Chemistry
- 10th: Latin 2
- 9th: Latin 1
- 9th: AP Biology
- 9th: Advanced Literature
- 9th: Exponents/Log/Trig A & B
- 9th: Euclidean Geometry 3
- 9th: Number Theory 2
- 9th: Graph Theory 1
- 9th: Studio Art: Paper

## 4 Years of Advanced Math Curriculum

| Year | Block 1               | Block 2               | Block 3                                  | Block 4                                  | Block 5             |
|------|-----------------------|-----------------------|------------------------------------------|------------------------------------------|---------------------|
| 9th  | Graph Theory          | Exp/Logs/Trig A       | Geometry 3                               | Exp/Logs/Trig B                          | Number Theory 2     |
| 10th | Discrete Probability  | Differential Calculus | Integral Calculus                        | Series and Topics in Calculus            | Number Theory 3     |
| 11th | Combinatorics 2*      | Linear Algebra A*     | Linear Algebra B*                        | Mathematical Logic*                      | Cryptography*       |
| 12th | Theory of Partitions* | Group Theory*         | Statistics A / Multivariable Calculus A* | Statistics B / Multivariable Calculus B* | Special Relativity* |

2017.08 - 2023.06

## Middle School

Portola Valley, California



Woodland School

3rd to 8th Grade

- Capstone Project: Hydroponics
- Mathcounts, AMC 8, and STEM Fairs

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## Experience

2023.07 - 2027.06

San Francisco, California



### Robotics Intern

Watney Robotics

*Assemble robot arms and controllers that allow humans from halfway around the world to teleoperate a robot; perform data labeling and ML models training.*

- Robot Arm Controller Assembly
- Teleoperation Testing
- Data Labeling
- ML Models Training

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## Volunteering

2022.07 - 2022.07

Portola Valley, California



### Camp Counselor

Woodland Summer Adventures

*Supervised students and gained first hand experience teaching small groups of students a new skill.*

- Leadership Development
- Safety Training
- CPR, AED & Basic First Aid Certified



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## Projects

2025.01 - 2025.12



### Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms

*Research Purpose: run Monte Carlo simulation to recreate evolutionary events inside a computer and see how the first digital worm evolves in silico to become the 'first fish' with a dorsoventrally inverted body plan after a somatic twist.*

- Computational Biology
- OpenWorm
- Sybernetic

2023.01 - 2023.03



### The Cliffhanger Project

*A novel mountain climbing aid that brings a new mechanical advantage to mountaineers by means of a simple lever mechanism — inspired by a physics demo harnessing a setup of three toothpicks to carry a payload hanging off the edge of a table.*

- Mountain Climbing Aid
- Mechanical Advantage
- Type 2 Lever

[General](#)[Education](#)[Experience](#)[Volunteering](#)[Projects](#)[Science fairs](#)[Contests](#)[Posters](#)[Papers](#)[Courses](#)[Societies & clubs](#)[Skills](#)[Languages](#)[Assessments](#)

## Science fairs

2025.03



### 4th Place, Biostatistics and Computational Biology

High School Division, Alameda County Science and Engineering Fair (ACSEF)

*Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms*

2024.04



### Honorable Mention, Zoology

Senior Division, California Science and Engineering Fair (CSEF)

*Why Do Vertebrates Have Decussated Corticospinal Tracts?*

2024.03



### 1st Place & Bill Tobin Award, Life Sciences

9th Grade, Golden Gate STEM Fair (GGSF)

*Why Do Vertebrates Have Decussated Corticospinal Tracts?*

[General](#)[Education](#)[Experience](#)[Volunteering](#)[Projects](#)[Science fairs](#)[Contests](#)[Posters](#)[Papers](#)[Courses](#)[Societies & clubs](#)[Skills](#)[Languages](#)[Assessments](#)**2023.04****Honorable Mention, Applied Mechanics and Structures**

Junior Division, California Science and Engineering Fair (CSEF)

*Cliffhanger: A Dynamically Stable Mechanical Mountain Climbing Aid by Design*

**2023.03****1st Place, Engineering: Electrical and Mechanical**

8th Grade, Golden Gate STEM Fair (GGSF)

*Cliffhanger: Dynamic Stability in a Mechanical Mountain Climbing Aid by Design*

**2023.03****1st Place, Engineering, Technology and Application of Science**

8th Grade, San Mateo STEM Fair

*Cliffhanger: Dynamic Stability in a Mechanical Mountain Climbing Aid by Design*

**2022.03****Honorable Mention, Biological Systems**

7th Grade, San Mateo STEM Fair

*Does the Color of Light Affect Plant Growth?*



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## Contests

2025.03



### Honorable Mention

USA Biolympiad (USABO)

*The most prestigious biology education and testing program for US high school students, participated by approximately 10,000 talented students annually.*

2024.04



### Maxima Cum Laude (Silver Recognition)

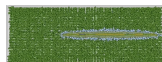
National Latin Examination

*Annual recognition of student accomplishment in Latin language learning and the study of Greco-Roman culture.*

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## Posters

2025.06.28

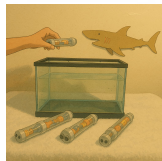


### Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms

25th International Worm Meeting, UC Davis

*Progress report on design aspects of evolutionary modeling for a digital worm and computational challenges encountered.*

2025.02.28



### A Twist on Decussation

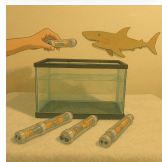
Junior Science and Humanities Symposium, San Francisco State University

*A comparative study of morphological models from ancient marine worms to the first fish inside an aquarium.*

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## Papers

2025.02.28



### Why Do Vertebrates Have Decussated Corticospinal Tracts?

Junior Science and Humanities Symposium, San Francisco State University

*A comparative study of morphological models from ancient marine worms to the first fish inside an aquarium.*

2025.01.25

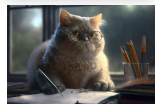


### The Rise of Homo Roboticus

Thesis Submission, Masason Foundation

*10 years later, how will artificial intelligence be further developed and what kind of changes will it bring to the world?*

2023.06.30



### People Get Essays Wrong

Global Essay Competition, John Locke Institute

*What is something important, about which nearly everybody is wrong?*



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## Courses

 The Structure and Interpretation  
of Computer Programs

UC Berkeley Pre-College Scholar

2025-  
06

 The Mathematics of Competitive  
Behavior

Johns Hopkins Center of Talented Youth

2024-  
08

 Engineering and Robotics  
Workshop at MIT Campus

National Geographic Student Travel

2024-  
07

 The Concord Review History  
Camp

The Concord Review

2024-  
06

 Logic: Principles of Reasoning

Johns Hopkins Center of Talented Youth 2023-07

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## Societies & clubs

### 🧬 Genetics Society of America

(2025–present) K-12 member of an international community of 6,000 biologists from 50 countries advancing the field of genetics.

### 🏛️ Berkeley Math Circle

(2015–present) Intermediate level II member; attend evening math lectures by professors at UC Berkeley campus every Wednesday.

### 🧑‍🦿 Benchmark Climbing

(2023–present) Member of afterschool climbing club; practice 2 to 3 times a week at an indoor bouldering gym in San Francisco.

### 🎸 Clock Tower Music

(2023–present) Taking guitar lessons with a local musician afterschool; practice new songs in the evenings.

### 🏋️ 24 Hour Fitness

(2024–present) Member of a neighborhood gym where I workout regularly in the evenings and on weekends.

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## Skills



### Hardware

Robot Arm Controller Assembly

Circuit Board Soldering

PC Assembly



### DIY

Hydroponics

3D Printing

Origami



### Software

Python

Data Labeling

ML Models Training



### Investment

Trading Strategies

Trading Tools

Risk Management

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## Languages



Latin

learning (2 years)



Japanese

limited phrases



Mandarin

limited phrases



Spanish

conversational (6 years)



English

native speaker fluency



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## Assessments

### SAT (10th Grade)

Score: 1510 (730 EBRW, 780 Math)

### SCAT (8th Grade)

CTY High Honors (484 Verbal, 501 Math)

### STB (8th Grade)

CTY High Honors (Score: 625)

### WISC-V (2nd Grade)

Extremely High (138 Full Scale, 146 Verbal)

## References

### Dr. Michael Yetman

Director of Science Curricula and Programs,  
Proof School

### Professor Zvezdelina Stankova

Founder and Director, Berkeley Math Circle, UC  
Berkeley



## ADAM CHEONG

2025: "The Structure & Interpretation of Computer Programs" (CS 61A), UC Berkeley Pre-College Scholar  
2025: 4th Place, Biostatistics & Computational Biology, Alameda County Science & Engineering Fair  
2025: Honorable Mention, USA Biolympiad (USABO)  
2024: "The Mathematics of Competitive Behavior", Johns Hopkins Center of Talented Youth  
2024: Maxima Cum Laude (Silver Recognition), National Latin Examination

2024: Honorable Mention, Zoology, California Science & Engineering Fair  
2024: 1st Place & Bill Tobin Award, Life Sciences, Golden Gate STEM Fair  
2023: "Logic: Principles of Reasoning", Johns Hopkins Center of Talented Youth  
2023: Honorable Mention, Applied Mechanics & Structures, California Science & Engineering Fair  
2023: 1st Place, Engineering: Electrical & Mechanical, Golden Gate STEM Fair  
2023: 1st Place, Engineering, Technology & Application of Science, San Mateo STEM Fair  
2022: Honorable Mention, Biological Systems, San Mateo STEM Fair

I've always loved science. I took biology in freshman year and scored a five on the AP Biology exam. As a sophomore, I am pursuing a "double science" track with AP Chemistry and AP Physics C: Mechanics, along with AP Calculus BC. I have also been a member of the Berkeley Math Circle for 10 years, where I have steadily developed my mathematical reasoning skills.

I am always excited by the challenge of applying what I learn to real-world problem-solving. I am a "founding intern" at Watney Robotics, a startup in San Francisco, where I assemble robot arm controllers for teleoperation, perform data labeling, and conduct ML models training. I aspire to identify and formulate the most important unsolved problems in robotics for our generation, and to create a future in which collective intelligence and resources are mobilized to solve them.

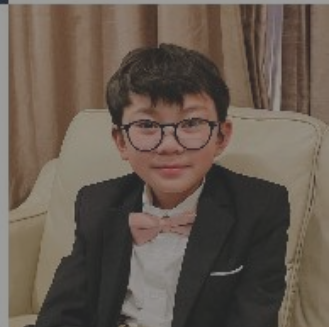
Alumni

KAITO  
NAKAJIMA

BORN IN 2009

Certified in 2018

#cross culture / language  
#mathematics



Generation 1

TAIHO HIGAKI  
BORN IN 2008

[READ MORE](#)

#AI / IoT  
#science

Generation 5

AKIRA  
HIRABAYASHI  
BORN IN 2009

COMING SOON

#chemistry  
#mathematics  
#statistics / data

Generation 3

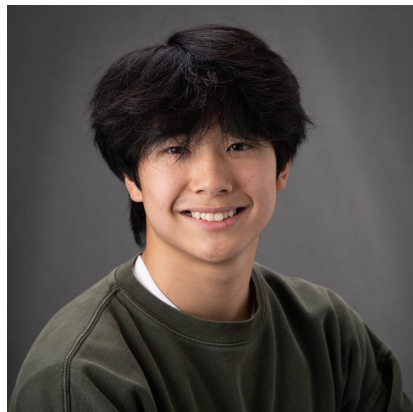
3期生

梶田 光

BORN IN 2008

READ MORE

#アート・音楽  
#プログラミング  
#数学



## ADAM CHEONG

2025年「The Structure & Interpretation of Computer Programs (CS 61A)」、カリフォルニア大学バークレー校・プレカレッジ・スカラー  
2025年 アラメダ郡科学技術フェア、生物統計学・計算生物学部門 第4位  
2025年 米国生物学オリンピック (USABO) 努力賞 (Honorable Mention)  
2024年「競争行動の数学」、ジョンズ・ホプキンス大学・才能ある青少年センター (CTY)  
2024年 全国ラテン語試験 マキシマ・ウム・ラウデ (銀賞)  
2024年 カリフォルニア州科学技術フェア、動物学部門 努力賞

2024年 ゴールデンゲート STEMフェア、生命科学部門 第1位 & ビル・トービン賞 受賞  
2023年「論理学: 推論の原理」、ジョンズ・ホプキンス大学・才能ある青少年センター (CTY)  
2023年 カリフォルニア州科学技術フェア、応用力学・構造部門 努力賞  
2023年 ゴールデンゲート STEMフェア、電気・機械工学部門 第1位  
2023年 サンマテオ STEMフェア、工学・技術・科学応用部門 第1位  
2022年 サンマテオ STEMフェア、生物システム部門 努力賞

私は幼い頃から、ずっと科学が大好きです。高校 1年生で生物を履修し、AP Biologyでは5を取得しました。現在は 2年生として、「ダブルサイエンス」として AP ChemistryとAP Physics C: Mechanicsに加え、AP Calculus BCも履修し、より深い学びへと進んでいます。また、Berkeley Math Circleには10年間在籍し、数学的思考力を着実に培ってきました。

私は常に、学びを現実の問題解決に応用することに強い関心と喜びを感じています。その思いから、サンフランシスコのスタートアップ Watney Robotics にて「創業期インターン」として、ロボットアームや遠隔操作コントローラーの組み立て、データラベリング、機械学習モデルのトレーニングなどに取り組んでいます。

そして私は、私たちの世代におけるロボティクスの最も重要な課題を常に見極め、その解決に向けて英知と資源が集集できる未来を継続的に築いていくことを信念としており、今後もその実現に向けて全力で取り組んでまいります。

5期生

大塚 蓮

BORN IN 2008

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#ロボット  
#生物・バイオ  
#科学

3期生

SAMAIRA MEHTA

BORN IN 2008

READ MORE

#コンピュータ  
#ビジネス  
#教育

8期生

# Profile Summary

| 1. | Experience「経験」                                       |                                                                                   | Technical Skills「技術力」                                                                                                 | Specialty「専門性」         |
|----|------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------|
| A. | <b>Watney Robotics</b><br>(Silicon Valley Startup)   |  | Robot Arm Controller Assembly,<br>Teleoperation, Data Labeling, ML Models<br>Training, Hydroponics, PCB Soldering     | Robotics               |
| B. | <b>Monte Carlo Simulation</b><br>(Open Science)      |  | OpenWorm, Sibernetica, <i>C. Elegans</i> , Monte<br>Carlo Method, Decussation, Somatic Twist,<br>Evolutionary Biology | Computational Biology  |
| C. | <b>Cliffhanger Project</b><br>(Mechanical Invention) |  | Mountain Climbing Aid, Mechanical<br>Advantage, Type 2 Lever Design,<br>Bouldering, Rock Climbing                     | Mechanical Engineering |

| 2. | Talented「異能」 Outstanding「優れている」                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3. | Wish to Achieve「これから成し遂げたいこと」                         |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------|
| a. | 25th International Worm Meeting: Poster Session                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | a. | 18 Math Courses on High School Transcript (Jun 2027)  |
| b. | 2 National Contest Awards:                                                                                                                                                                                                                                                                                                                                                                                                                      | b. | Research Paper for Regeneron STS (Jun 1~Nov 1 2026)   |
| c. | 7 STEM Fair Prizes:        | c. | 7 AP STEM Exams: 4 Science, 2 Math, 1 CS (Jun 2026)   |
| d. | WISC-V: Extremely High (Full Scale IQ: 138)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | d. | UC Berkeley Course (4 Credits): CS 61A (Jun~Aug 2025) |